

Neuron Modality Labels & Hallucination Enrichment

PMBT, $p < 0.001$, min-100/max-300 descriptions

MH-Neuron / Modality Taxonomy

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1. Modality-label distribution (PMBT · gate)

A neuron is *unknown* if it has < 5 high-activation tokens; otherwise *visual/text* if its visual-vs-text activation-rate difference is significant ($p < 0.001$), else *multimodal*.

Model	Neurons	Visual	Text	Multimodal	Unknown
LLaVA-1.5-7B	352,256	111,893 (31.8%)	205,872 (58.4%)	34,456 (9.8%)	35 (0.0%)
LLaVA-OneVision-7B	530,432	249,570 (47.1%)	180,289 (34.0%)	75,489 (14.2%)	25,084 (4.7%)
InternVL2.5-8B	458,752	157,956 (34.4%)	214,604 (46.8%)	86,159 (18.8%)	33 (0.0%)
Qwen2.5-VL-3B	396,288	173,234 (43.7%)	129,295 (32.6%)	41,396 (10.4%)	52,363 (13.2%)

2. Hallucination-driving enrichment by ΔH — odds ratios

Driving neurons = top 5% by ΔH (change in hallucination rate when the neuron is ablated); Fisher's exact test per category. OR > 1 = enriched $\uparrow$, OR < 1 = depleted $\downarrow$; **bold** = significant ($p < 0.05$) and ≥ 1.5 or ≤ 0.67 . POPE = visual hallucination; TriviaQA = text knowledge. The expected double dissociation: visual enriched on POPE / depleted on TQA, text the reverse. Only models/combo with results on disk are shown.

POPE (visual hallucination)

Model	Combo	Visual	Text	Multimodal	Unknown
LLaVA-1.5-7B	PMBT·gate	0.93 $\downarrow$	1.12 $\uparrow$	0.87 $\downarrow$	0.00 $\downarrow$
LLaVA-1.5-7B	PMBT·gate_up	1.04 $\uparrow$	1.05 $\uparrow$	0.80 $\downarrow$	0.25 $\downarrow$
LLaVA-OneVision-7B	PMBT·gate_up	0.36 $\downarrow$	2.15 $\uparrow$	1.77 $\uparrow$	0.04 $\downarrow$
InternVL2.5-8B	PMBT·gate	0.48 $\downarrow$	1.81 $\uparrow$	0.98 $\downarrow$	22.82 $\uparrow$
InternVL2.5-8B	PMBT·gate_up	2.11 $\uparrow$	0.62 $\downarrow$	0.71 $\downarrow$	0.00 $\downarrow$
InternVL2.5-8B	FT·gate	0.62 $\downarrow$	0.87 $\downarrow$	1.40 $\uparrow$	2.07 $\uparrow$
Qwen2.5-VL-3B	PMBT·gate	2.02 $\uparrow$	0.81 $\downarrow$	1.04 $\uparrow$	0.03 $\downarrow$
Qwen2.5-VL-3B	PMBT·gate_up	1.38 $\uparrow$	0.91 $\downarrow$	0.93 $\downarrow$	0.02 $\downarrow$
Qwen2.5-VL-3B	FT·gate	1.18 $\uparrow$	0.78 $\downarrow$	0.93 $\downarrow$	1.06 $\uparrow$

TriviaQA (text knowledge)

Model	Combo	Visual	Text	Multimodal	Unknown
LLaVA-OneVision-7B	PMBT·gate_up	0.94 $\downarrow$	1.01 $\uparrow$	1.15 $\uparrow$	0.62 $\downarrow$
InternVL2.5-8B	PMBT·gate	1.52 $\uparrow$	0.74 $\downarrow$	0.83 $\downarrow$	0.00 $\downarrow$
InternVL2.5-8B	PMBT·gate_up	1.46 $\uparrow$	0.82 $\downarrow$	0.82 $\downarrow$	0.21 $\downarrow$
InternVL2.5-8B	FT·gate	1.29 $\uparrow$	0.69 $\downarrow$	1.02 $\uparrow$	0.27 $\downarrow$
Qwen2.5-VL-3B	PMBT·gate	0.88 $\downarrow$	1.21 $\uparrow$	0.85 $\downarrow$	1.03 $\uparrow$

Coverage / caveats. LLaVA-OneVision-7B and LLaVA-1.5-7B currently have *gate_up only* — their gate and FT step-10 jobs are still running; LLaVA-1.5 is the validation model and runs POPE only (no TriviaQA). Qwen2.5-VL-3B TQA is gate-only so far. Per-neuron ΔH is quantised (batch-of-50 ablation), so the top-5% cutoff often lands on a tie at the boundary — treat marginal ORs cautiously — but the ΔH ranking is the paper’s headline and reproduces the visual/text double dissociation. Labels from `llm_permutation_min100_max2048/` (native $\alpha = 0.001$ on max-300 text for these models).